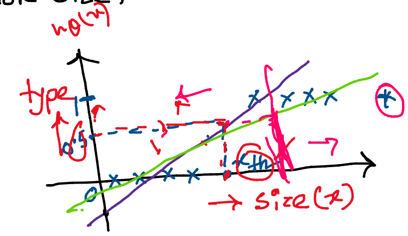
Friday, April 1, 2022 10:44 PM

Classification:

'y' is discrete e.g. in binary classification 'y' can only have two values: {0, 1}

x_1	x_2	x_3	y
3	0.2	24	0
6	0.5	40	1

Let's say, given only the tumor size, we need to predict whether it is benign or malignant



$h_\theta(x) = \theta^T x = \theta_0 + \theta_1 x$

→ $h_\theta(x) \geq 0.5$ → malignant $\equiv x \geq x_{th}$

→ $h_\theta(x) < 0.5$ → benign $\equiv x < x_{th}$

→ Problems in using Linear regression for classification:

(1) Sensitive to outliers

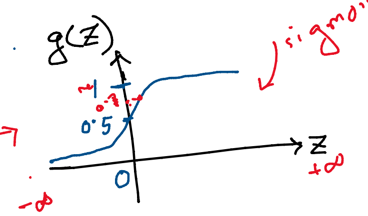
→ (2) $y = 0$ or 1 but $h_\theta \leq 0$, $h_\theta \geq 1$

$h_\theta(x) = \theta^T x$ → $h_\theta(x) = g(\theta^T x)$

Solution: (Logistic regression)

Idea → Transform $\theta^T x$ such that $0 \leq h_\theta(x) \leq 1$

Transformation function, $g(z) = \dfrac{1}{1+e^{-z}}$



$$\hat{y} = \begin{cases} 1 ; & \text{if } h_\theta(x) \geq 0.5 \\ 0 ; & \text{if } h_\theta(x) < 0.5 \end{cases}$$

$$\text{or, } \hat{y} = \begin{cases} 1 ; & \text{if } \dfrac{1}{1+e^{-\theta^T x}} \geq 0.5 \\ 0 ; & \text{if } \dfrac{1}{1+e^{-\theta^T x}} < 0.5 \end{cases}$$

interpretation:

$h_\theta(x) = P(y=1 \mid \theta, x)$ → Probability of y=1

Let's say, we have determined the optimum value of "θ" for our model as:

1d

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix} = \begin{bmatrix} 0.54 \\ 0.1 \end{bmatrix}, \quad x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

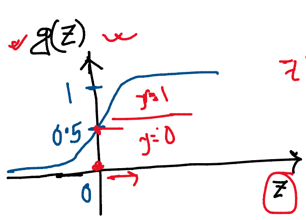
Now, $\theta^T x = \begin{bmatrix} 0.54 & 0.1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = 0.54 + 0.3 = \boxed{0.84}$

So, $g(\theta^T x) = \dfrac{1}{1+e^{-0.84}} = \boxed{0.7}$ → 70% probability that y=1 or the tumor is malignant

Decision boundary:

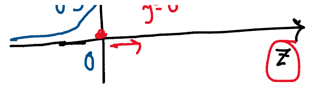
$h_\theta(x) = \dfrac{1}{1+e^{-\theta^T x}}$

$y = \{ 1, h_\theta(x) \geq 0.5$ → $z \geq 0$



$z = \theta^T x$

So, $z = \theta^T x = 0$ is the decision boundary!

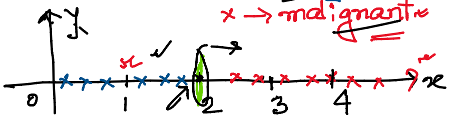
$1+e^{-z}$  **boundary!**

$$y = \begin{cases} 1, & h_\theta(x) \geq 0.5 \quad (z \geq 0) \\ 0, & \text{otherwise} \quad (z < 0) \end{cases}$$

*. Example-1:

$x \to$ benign

$x \to$ malignant



$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ x \end{bmatrix}$$

Now, $\theta^T x = \theta_0 + \theta_1 x = -2 + x$

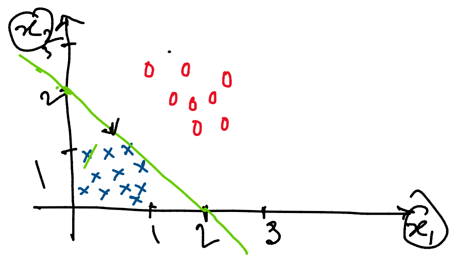
Decision boundary,

$\theta^T x \geq 0$

or, $x - 2 \geq 0$

or, $\boxed{x \geq 2}$

*. Example-2:

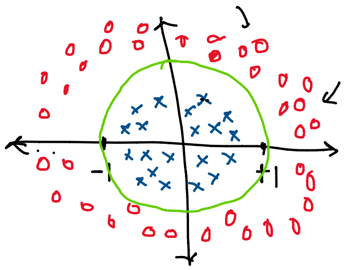


$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

Decision boundary, $\theta^T x \geq 0 \Rightarrow -2 + x_1 + x_2 \geq 0$

$\Rightarrow x_1 + x_2 \geq 2$

*. Example-3:



$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_1^2 \\ x_2^2 \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

Decision boundary, $\theta^T x \geq 0 \Rightarrow x_1^2 + x_2^2 \geq 1$

⊞. Finding the model ≡ Finding the best "θ" [Learning Algorithm]

$$D = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots\ldots (x^{(n)}, y^{(n)})\}$$

$$\rightarrow x^{(i)} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_d \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_d \end{bmatrix}$$

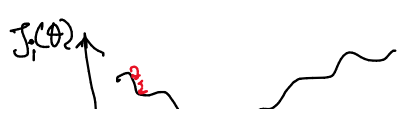
$$\rightarrow \boxed{h_\theta(x) =} g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

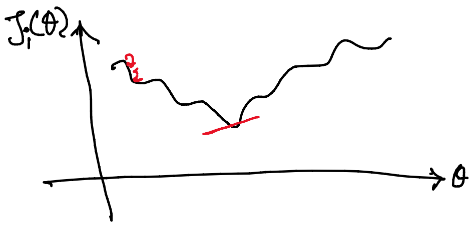
Task: Find the best "θ" that minimizes the cost function.

$\boxed{\text{Option-1}}$

Let's define our cost function as MSE: $J(\theta) = \frac{1}{n} \sum_{i=1}^{n} J_i(\theta)$

Now, $J_i(\theta) = [h_\theta(x) - y]^2 = \left[\frac{1}{1 + e^{-\theta^T x}} - y\right]^2$





Option-2

*. Binary crossentropy loss:

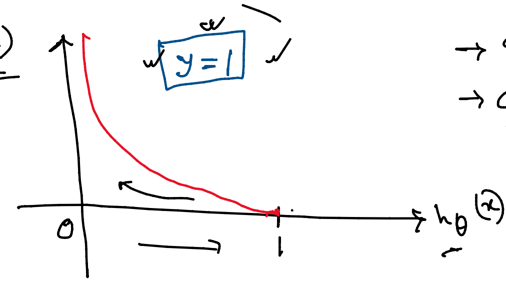
$$J_i(\theta) = \begin{cases} -\log(h_\theta(x)) & \text{if } y=1 \\ -\log(1-h_\theta(x)) & \text{if } y=0 \end{cases}$$

$\log(0)$

$$\text{or, } J_i(\theta) = -\left[y \times \log(h_\theta(x)) + (1-y)\log(1-h_\theta(x))\right]$$

Interepretation:

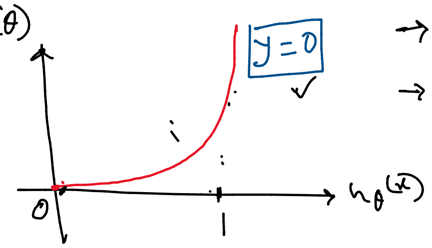
① $J_i(\theta)$



→ cost = 0 if $y=1$ and $h_\theta(x)=1$

→ cost = ∞ if $y=1$ and $h_\theta(x)=0$

② $J_i(\theta)$



→ cost = 0 if $y=0$ and $h_\theta(x)=0$

→ cost = ∞ if $y=0$ and $h_\theta(x)=1$

*. Optimizer:

Gradient descent:

θ^0 = random

while not converged:

$\theta^{k+1} = \theta^k - \alpha \nabla_\theta J$

if $\nabla_\theta J \approx 0$:

converged = True

Notation for logistic regression: